

Fig. 1: Circuit for movie maker

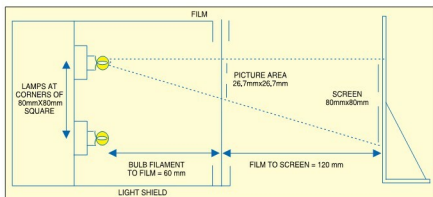


Fig. 2: Assembly arrangement

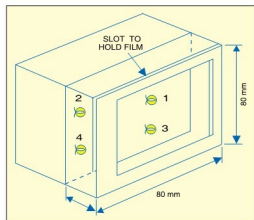


Fig. 3: Plastic case with assembled circuit

using switch S1 and make sure that the outputs of IC2 (0 through 3) are normally low but briefly go high three-four times within a second. Also ensure that the lamps flash one at a time in a repeating sequence. If the sequence appears to be wrong or any of the lamps fails to glow, check the wiring.

The light shield and film holder can be made of a thin card, sheet metal or plywood. Strictly adhere to the various dimensions as shown in Fig. 2. Otherwise, the shadow images may fail to register properly when projected.

Use a plastic cabinet as shown in Fig. 3 to hold the circuit board and battery. Owing to the power requirements of the lamps, it is more economical to use four 1.5V cells in a battery box. Else, you can use a 6V power adaptor.

There are two ways of mounting the lamps. The more satisfactory but more expensive method is to bolt the four lamps. Alternatively, drill four 1cm dia. holes on the front of the cabinet, wedge the base of the lamps in these holes and solder wire to the bases.

Fig. 4 shows an example of film

making. The easiest way to prepare the film frames is to photocopy the desired drawings onto transparent films. Alternatively, trace them on a transparent acetate film or draughtsman's film, using a fine marker pen. Align all the drawings on the frames

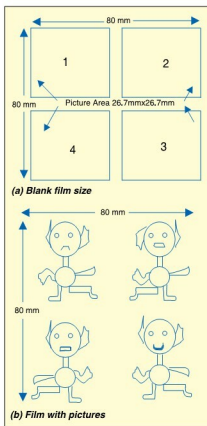


Fig. 4: Film making

and project onto the screen.

Working of the circuit is simple. First of all, fix the clock frequency at about 16 Hz. Place the film on the holder. Ensure a distance of 12 cm between the screen and the assembled unit and power-on the circuit using switch S1. Now you can see your drawings as a short movie clip on the screen

EFY note. We have tested this circuit without the mechanical arrangement. ●